Causal Feature Engineering and Learned Residual Evidence for Online Structural Break Detection

Author: Koen Engels

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Abstract

The Abu Dhabi Investment Authority (ADIA) supports the independent research institute ADIA Lab. I study causal structural-break detection in its Real Time Challenge with CrunchDAO. The task requires a score after every online observation and evaluates cross-sectional ranking using time-stratified area under the receiver operating characteristic curve (TS-AUC).

My selected submission combines engineered statistical features, autoregressive residual representations and a Light Gradient Boosting Machine (LightGBM) student trained on nested cross-fitted soft labels. It achieved a recorded public leaderboard score of 0.6191.

Subsequent offline research examines learned normal-process models, neural and pretrained forecasting representations, nested width selection, temporal blending and sequential residual evidence. On matched search folds, a combination using a selected-width gated recurrent unit (GRU) null model scores 0.622489, and adding sequential mean, energy and lag-product evidence raises the mean to 0.623224. The incremental gain is positive in nine of fifteen fold–seed cells, but fails all five two-half bootstrap gates for every tree seed against the preceding combination. Temporal analysis locates most of the additional gain before online step 100, with a regression between steps 100 and 249. A subsequent 8,000-series likelihood search for generalized autoregressive score models with Student-t innovations (GAS-t) raises its matched blend from 0.619810 to 0.620676, but fails all 105 declared gates and remains below the GRU combinations.

I document scorer and leakage corrections, distinguish search evidence from fixed-holdout confirmation, and report unsuccessful methods alongside promising candidates. As of 28 October 2026, submission 46 remains the selected recorded public result; the competition run window is closed, and neither sequential residual evidence nor the GAS-t blend has received a new public score or a registered holdout confirmation. The results support complementary residual representations as a research direction, while leaving replacement of the selected submission unconfirmed.

Keywords: online structural break detection; time-stratified AUC; feature engineering; nested cross-fitting; soft labels; learned null models; stacking; sequential evidence; cluster bootstrap.

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1 Introduction

Structural changes can invalidate a forecasting or monitoring model without producing a single extreme observation. A detector must identify sustained changes in location, scale or dependence while retaining sensitivity to weak transitions. The difficulty increases when observations arrive sequentially and the change location is unknown.

ADIA Lab is an independent institute supported by the Abu Dhabi Investment Authority (ADIA). Its challenge with CrunchDAO in 2026 imposes this online setting: historical observations are available initially, but each online score must be returned before the next observation is revealed [1]. Its mechanics differ from the 2025 competition, where both segments were available and the candidate boundary was given [2]. Consequently, receiver operating characteristic area under the curve (ROC-AUC) figures from the previous competition cannot be used as numerical baselines for the current TS-AUC task.

Previous competition write-ups illustrate the value of diverse statistical features and ensembles [3]. My report follows that technical solution-report form but addresses the online constraint. I contribute: (i) an audited account of the selected feature-based submission; (ii) matched comparisons of learned-null and foundation-model routes; (iii) a causal sequential-residual representation evaluated with nested feature construction; and (iv) a failure analysis separating average improvements from statistical acceptance. I do not claim a new globally optimal detector or an exhaustive architecture benchmark.

1.1 Related work and the previous edition

The 2025 winning-team solution report [3] and the previous-edition task documentation [2] provide comparative context. The winning report combines several statistical feature representations through tree-based stacking. It motivates examining complementary representations, but its known-boundary, complete-segment setting differs from causal monitoring at an unknown change location. I therefore discuss its architecture rather than using its scores as baselines. This comparison does not establish that every previous-edition paper was reviewed or that those publications guided all earlier experiments.

2 Problem formulation and data

2.1 Data supplied by the organizers

I received the ADIA Lab–CrunchDAO Real Time dataset, described in the official dataset documentation [1]. The files are obtained through the competition Hub and its participant setup. My local copy contains 10,000 labelled training series and a separate labelled reduced test set of 100 series. The full public and private test sets each contain 10,000 series and are evaluated on the platform; their labels are unavailable to my inference code.

The organizers describe a mixture of real-world and synthetic data-generating processes (DGPs), with possible changes in location, scale, distributional shape or dependence. The supplied files do not identify each series' exact DGP, original source domain or whether it is real or synthetic. This limits what I can infer about the collection: financial and sensor processes are possible interpretations, not established provenance. My retrospective break-type analysis is a diagnostic constructed from observations, not organizer-supplied DGP metadata.

Each series has a break-free historical reference, typically 1,000–5,000 observations, followed by an online segment of 10–1,000 observations. A series has at most one break, located somewhere in the online segment, or no break. Values are standardized separately for each series using its historical mean and standard deviation; those normalization parameters remain fixed during online inference [1]. In the supplied Parquet files, the series identifier and observation index form an `(id, time)` index, and `period` distinguishes history (`1`) from online observations (`2`).

Supplied local file	Contents and role
X_train.parquet	Observation values and period indicators for the 10,000 training series; indexed by id and time
y_train.parquet	Binary target for every online training observation: zero before a break, one from the break onward
y_train_index.parquet	Series-level break position: tau_index counts from zero within the online segment; a negative index denotes no break; tau is also supplied
X_test.reduced.parquet	Observations and period indicators for the 100-series local reduced test
y_test.reduced.parquet; y_test_index.reduced.parquet	Online labels and series-level break positions for that reduced test

Table 1. *Organizer-supplied training and reduced-test files used in this study. The file schemas were checked against the actual Parquet metadata. These are input data, distinct from the feature caches and retrospective analyses generated during my research.*

No forward-looking inference code is permitted: the detector must return its score for the current observation before receiving the next one, and the online iterator cannot be reread [1]. Consequently, a prediction may use the complete historical segment and the online prefix received so far, but no later online values, normalization over the complete online segment, or evaluation break labels. Training labels supervise fitting; retrospective break locations and types support evaluation only. My offline batch feature builders must reproduce this same prefix-only information restriction at every scored row.

2.2 Prediction target and evaluation metric

For series i, let H_i denote its historical segment and $x_{i,0:t}$ its observed online prefix. A score $s_{i,t}$ must be measurable with respect to these observations alone. With τ_i denoting the online change location, the binary target is:

$$y_{i,t} = 1\{\tau_i \text{ exists and } \tau_i \le t\}. \tag{1}$$

At each time t, the area under the receiver operating characteristic (ROC) curve (AUC), denoted AUC(t), compares positive and negative series that remain observable. The official pair-weighted metric is [4]:

$$\text{TS-AUC} = \Sigma_t\, n_+(t) n_-(t)\, \text{AUC}(t) \,/\, \Sigma_t\, n_+(t) n_-(t). \tag{2}$$

A time-only score cannot discriminate within a fixed t. Forecasting negative log-likelihood (NLL), pooled row AUC and time-stratified AUC therefore answer different questions. In particular, a change in a common time prior need not improve this ranking.

2.3 My research partition and derived diagnostics

From the supplied training data, I built a full causal feature cache containing 5,036,517 online rows; sampling every 25th online step yields 206,261 research rows. Historical lengths range from 1,000 to 5,000, with median 2,998. I reserve a fixed 20% of training series for confirmation and use the other 8,000 for search. This 2,000-series holdout is my own research partition, not the organizer's private test set. The separate 100-series reduced test had already been consulted during research.

Retrospective metadata classifies 4,866 break series by their dominant change: 2,572 variance, 1,938 autocorrelation and 356 mean changes. This taxonomy is computed using post-break observations and is used only for evaluation. It does not establish the original domains of the individual series. I neither assume a uniformly financial dataset nor provide retrospective break type, strength or age to the detector.

3 Methodology

3.1 Statistical features and the submission model

The feature bank combines expanding and rolling moments, level and scale contrasts, hypothesis-test statistics, tail indicators, cumulative sum (CUSUM) statistics, autoregressive (AR) changes, generalized autoregressive conditional heteroskedasticity (GARCH) states and generalized likelihood ratios (GLRs). Raw and residual-transformed streams are evaluated in parallel. Bayesian information criterion (BIC)-weighted autoregressive models remove predictable mean dynamics; additional variance and Rosenblatt and probability integral transform (PIT) representations provide alternative views.

The historical union152 bank combined raw, whitened, GARCH and GLR information. Removing context columns produced the 144-column training line used in the final student package. More columns were not uniformly beneficial: a 731-column bank lost to smaller nested banks. Likewise, leave-one-family-out importance did not measure a family's standalone value when several families substituted for one another.

A historical forward-selection audit illustrates the interaction: on a repeatedly used local reduced evaluation panel, raw84 scored 0.55170; adding whitened level features raised the local score to 0.62591; adding whitened fit features then raised it to 0.65767. The fit family alone did not improve raw84. **All three figures are local research scores, not public leaderboard results or independent confirmation.** The 104-feature configuration scoring 0.65767 was never publicly evaluated. The selected submission 46 achieved public TS-AUC 0.6191 with a different configuration and evaluation cohort, so subtracting these scores would not estimate its local-to-public generalization gap.

I used the Light Gradient Boosting Machine (LightGBM) [5] for the main feature classifier and a teacher–student training scheme to soften its binary targets. The teacher is an auxiliary classifier that estimates the probability of a break; the student is the detector trained on a mixture of those probabilities and the observed zero/one labels. This supplies graded training targets rather than treating every labelled row as equally certain. The teacher generates targets during training; the student produces the online predictions.

For each student training row, the teacher prediction $q_{i,t}$ is generated by inner series-level cross-fitting entirely within the outer training pool, so the teacher sees neither that row's series nor the outer validation series and their labels cannot leak into the student's training targets. Soft labels take the form:

$$\tilde{y}_{i,t} = (1 - \alpha) y_{i,t} + \alpha q_{i,t}. \tag{3}$$

Submission 46 used $\alpha=0.5$ and five student seeds. Although its package name refers to learning using privileged information [6], ablations did not establish a gain from the additional privileged teacher input. The evidence supported soft-label regularization. A later selected recipe, trial312, uses $\alpha=0.14595873$, ten leaves and extra trees, with 600 rounds in the matched offline evaluation.

3.2 Learned normal-process model

A two-layer gated recurrent unit (GRU) [7] predicts a one-step mean $\hat{\mu}_t$ and the logarithm $\hat{v}_t$ of the squared scale. Training minimizes a Student-t negative log-likelihood with $\nu=5$ in the frozen v5 recipe. Only normal training material is used: historical segments and, in applicable versions, labelled pre-break online portions of training series. Sampling is proportional to usable length.

$$L_t = \tfrac{1}{2}\hat{v}_t + \tfrac{1}{2}(\nu+1)\log[1 + (x_t - \hat{\mu}_t)^2 / (\nu \exp(\hat{v}_t))]. \tag{4}$$

$$z_t = (x_t - \hat{\mu}_t) / \exp(\tfrac{1}{2}\hat{v}_t). \tag{5}$$

Residuals are recentered and rescaled using the model's surprises on the same series' historical tail. Original null features comprise mean/variance and autocorrelation GLRs, mixture and maximum evidence, inferred evidence age, recent residual energy and historical/online likelihood summaries. A separate LightGBM maps these features to a null-derived break score.

The principal combination gives the student weight 0.8 and the null-derived score weight 0.2 for $t<500$; the student alone is used thereafter. Component scores are percentile-ranked in the local fold evaluations. These fold-relative ranks are part of the offline evaluation convention. Their portable replacement by training-fitted cumulative distribution functions (CDFs), and equivalence under strictly streaming inference, remain necessary checks for a new exported package. Causality of residual features alone does not establish export-equivalence of the blend.

3.3 Nested GRU width selection

For each outer fold, a fixed 25% subset of its training series forms a calibration panel. Widths 64, 128 and 256 are compared using downstream calibration TS-AUC. The selected widths across five outer folds are 128, 64, 128, 256 and 64. These are fold-specific choices, not evidence for a universal width of 256.

Null-classifier training features are generated by four inner cross-fitted GRUs; an outer-training GRU produces the validation features. Thus, label-informed normal-material selection never uses the scored outer series. The selected GRU checkpoints are shared by tree seeds 6, 7 and 8. The seeds vary the downstream fits, not the GRU training realization or the underlying dataset.

3.4 Temporal and multiscale extensions

I preregistered independent changes. Early-off replaces the combination with trial312 before $t=100$. Late-on retains the original early rule and chooses a late weight from $\{0,0.1,0.2,0.3\}$ using training-only calibration TS-AUC. Multiscale adds mean, energy, lag-product and likelihood summaries over causal windows of 10, 25, 100 and 250 observations.

A separate adaptive gate uses time, history length, historical null NLL, short/medium residual energy and disagreement between null and student scores. A bounded sigmoid weight is optimized on exact calibration TS-AUC, with regularization and a fixed-rule fallback. The gate is frozen before outer validation. It is evaluated separately from the other changes rather than stacked post hoc.

3.5 Sequential residual accumulation

The final extension operates on the calibrated GRU residuals. A clipping bound is the historical 99th percentile of absolute residuals, constrained to [1,10]. Three bounded statistics represent mean, squared energy and lag-1 residual product. The first online lag product uses the last historical residual.

For statistic T and a fixed tilt λ, history-only empirical normalization defines diagnostic log evidence:

$$\ell\lambda,t = \lambda T(z_t) - \log[(1/m)\,\Sigma_j \exp\{\lambda T(z^h_j)\}]. \tag{6}$$

$$\log R\lambda,t = \operatorname{logaddexp}(0, \log R\lambda,t-1) + \ell\lambda,t. \tag{7}$$

The initial log state is $-\infty$. Mean tilts are $\{-0.5,-0.25,0.25,0.5\}$; energy tilts are $\{-0.15,0.15\}$; lag-product tilts are $\{-0.25,0.25\}$. Equal mixtures combine the separate tilted states within each family. I output mixture level, running peak and slopes over 25 and 100 observations, giving twelve additional features. The classifier and fixed blend recipe remain unchanged.

The recursion is motivated by Shiryaev–Roberts (SR) detection [8]. Its theoretical optimality or false-alarm guarantees do not automatically apply: my normalization is empirical and residuals may be dependent. This is a diagnostic evidence representation, not a proved martingale construction.

3.6 Normality, distributional diagnostics and feature contribution

I distinguish Gaussian shape from an unchanged generating process. A normal-history GRU models the pre-break process; it does not assert Gaussian innovations. A stable heavy-tailed series can be non-Gaussian without a break, and a break can preserve Gaussian shape while changing its mean, variance or dependence. Normality is therefore a diagnostic feature, not the break label.

Expanding and rolling standardized moments include skewness and excess kurtosis. The implemented Jarque–Bera (JB) statistic [16] combines them with sample size:

$$JB_n = (n/6)[\text{skew}_n^2 + \text{excess-kurtosis}_n^2/4]. \tag{8}$$

The code uses population central moments, neutral higher moments for very short windows, and guards against cancellation when the mean is large relative to the scale. Its classical χ^2_2 reference is asymptotic under suitable independent sampling. Serial dependence, estimated residual transforms and repeated overlapping windows prevent me from treating an unadjusted p-value as a calibrated online false-alarm rate. No claim that all histories pass a normality test follows from these features, and no Shapiro–Wilk campaign is established by the saved evidence.

Historical quantile thresholds also support approximate Kolmogorov–Smirnov (KS), Cramér–von Mises and tail-weighted Anderson–Darling style deviations of the observed online distribution from its own history. These compare distributions; they are not all Gaussian goodness-of-fit tests. Student-t NLL, residual energy, autocorrelation and historical PIT/Rosenblatt transforms provide complementary model-based checks. An empirical normal-score transform can improve marginal

shape while retaining temporal dependence; it is not proof that the transformed observations are independent standard normals. The learned-null history-tail NLL is a fit diagnostic whose downstream TS-AUC must be measured separately.

I use several views of feature contribution: split gain and Shapley additive explanations (SHAP) for fitted-model usage, feature correlation for duplication, a SHAP interaction graph for potential non-additivity, elastic-net generalized linear model (GLM) coefficients for a simpler functional form, and refitted out-of-fold (OOF) additions/removals for predictive contribution. They answer different questions. A large split gain does not establish incremental value; a near-zero coefficient does not establish that a tree feature can be deleted; neither observational attribution nor an ablation identifies a causal effect in the original data-generating process.

The historical SHAP graph retained the top 2% of non-meta interaction edges: 21 edges formed one 14-feature component, with 74 singletons. Its component linked historical shape/dependence descriptors to long-window statistics. This describes the sampled fitted model at that threshold, not universal additivity. A coefficient diagnostic consequently added 20 explicit historical-shape × long-window products to the 84-feature GLM input. With five series folds and 1,502,019 sampled rows, four of the resulting 104 columns had very small mean absolute elastic-net coefficients. The stochastic gradient descent (SGD) fits also emitted maximum-iteration warnings; I therefore treat this as a diagnostic rather than a fully converged selection certificate.

3.7 Ablation matrix: small-to-large and large-to-small selection

For a feature set F and family G, denote the score from refitting the same learner on a fixed series split by S(F). The complementary inclusion and exclusion contrasts are:

$$\Delta\text{in}(G \mid F) = S(F \cup G) - S(F); \; \Delta\text{out}(G \mid F) = S(F) - S(F \setminus G). \tag{9}$$

Both are context-dependent. In small-to-large forward selection, I start with a sparse core and measure each candidate addition, then repeat after accepting a family. In large-to-small selection, I start from the full bank and refit after removing one family or a joint block. A family can contribute strongly to a sparse core and be replaceable in a saturated bank. Conversely, two individually dispensable families can become important when removed together. Grouped bootstrap stability and nested selection are needed before exporting a selected subset; the existence of a general framework does not establish that every stage was completed for every bank.

The historical reduced-panel matrix uses the same 600-round binary recipe and common raw84 baseline. Whitened level features add +0.074214 alone; fit features add −0.000708 alone but +0.031760 after level features. Starting from the full152 control, dropping fit features costs approximately 0.02355; dropping whitened CUSUM costs approximately 0.03328; dropping both costs approximately 0.03092. These nonlinear contrasts refute a simple independent ranking. Figure 3 shows the saved additions and available removals; blank cells are unmeasured comparisons, not zeros.

The greedy historical path stopped at raw84 + level_white + fit_white, 104 columns and reduced TS-AUC 0.65767; all remaining additions in the next round lost. This reused reduced panel was a research screen, not an independent confirmation or the final submitted bank. A separate grouped union144 drop-family campaign saved fold signs and gates: for example, removing long-window whitened features reduced pooled OOF score by 0.002257 but did not establish significant harm; removing both tail representations reduced it by 0.003064 and lost on all five folds. Its pooled OOF protocol differs from the historical reduced screen, so I do not merge these scores into one homogeneous matrix.

A later family-score forward build investigated a different selection unit: one learned score per family, rather than the family's original columns. Its construction, results and nesting limitations are detailed below. Complementarity among raw columns and complementarity among family predictions are distinct selection problems.

3.7.1 Family mini-models: one learned feature per family

I trained a small classifier on each feature family and used its break probability as a single new feature. For example, a model receiving only whitened long-window mean, scale and dependence statistics produced one long-window family score. A separate model compressed tail statistics into one tail score. This is supervised compression: the score is learned from training labels, rather than being a simple average of the family's columns or its first principal component. The second-stage learner receives these scores, rather than the complete original feature bank:

$$u^{g}_{i,t} = f^{g}(X^{g}_{i,t}); \; s_{i,t} = h(u^{g1}_{i,t}, \ldots, u^{gK}_{i,t}, t). \tag{10}$$

Here X^{g} contains family g's causal features, f^{g} is its mini-model and h is the combiner. Thus one family becomes one model-derived input feature; this is a family-level stacking experiment. The online-time input lets the combiner change how it uses evidence over time, even though time alone cannot discriminate within a time bucket.

The saved implementation evaluated 24 families: 18 raw/whitened statistical groups, the v5 learned null, Allan and PIT-Allan summaries, detector-score velocity, a Kalman model, and Bolt embeddings. Each mini-model was a binary LightGBM with 15 leaves, at least 200 rows per leaf, learning rate 0.05 and 250 rounds. In each of five outer series folds, four inner folds generated mini-model predictions for combiner training; refitting each mini-model on the outer training pool generated its outer validation inputs. This prevents a family's classifier from supplying its own in-sample training predictions to the final combiner.

Greedy selection started with the null score. Candidate scores with absolute Spearman correlation at least 0.8 to a selected score were skipped. A two-fold series split inside the outer training pool evaluated additions with a 150-round combiner; selection stopped below a gain of 0.0005 or at twelve families. The final combiner used 300 rounds, and outer scores were saved for every prefix of the training-selected order. The terminal prefix, rather than the best outer prefix, defines the result below. Four to six family scores were retained per fold; whitened long- and medium-window families were added in all five folds. Bolt was selected once, which establishes occasional selection rather than a confirmed contribution to the full model.

Outer fold	Retained family scores	Compact model TS-AUC	Matched submission-46 control	Delta
1	4	0.60331	0.61592	-0.01261
2	5	0.61096	0.62910	-0.01814
3	6	0.60979	0.61827	-0.00848
4	6	0.61030	0.61782	-0.00752
5	5	0.60547	0.61393	-0.00846

Table F1. *Historical terminal-prefix scores: mean 0.607966 versus 0.619008, a loss of 0.011042; every fold lost. This cohort differs from the later 8,000-series search campaign. Compact representation was not an accepted replacement; the script supplies no separate five-bootstrap-seed acceptance result. Sources: src/family_forward_build.py, artifacts/family_forward_build.json and family_minimodel_fold_source.csv.*

The scope is narrower than "fully nested" might suggest. The family classifiers were cross-fitted, but upstream null and embedding tables were loaded from precomputed caches; this script does not establish their complete outer-fold independence. The two-fold selection scorer reused the four-fold family predictions without refitting those family learners inside each selection split, so its inner selection scores can be optimistic. The loader also does not exclude the fixed research holdout: this historical grouped training-data screen is not a search-only run or a new holdout confirmation. I therefore report its implementation and negative result without certifying leakage-free independence of the entire hierarchy. The loss is consistent with compression discarding useful within-family interactions or an inadequate capacity/selection recipe; the experiment does not isolate which explanation caused it.

3.8 GLMs, pairwise ranking and selection objectives

Two logistic analyses must be distinguished. The ordinary GLM predicts the row label using standardized features and explicit interaction products, fitted with elastic-net SGD. A historical blend used 85% LightGBM and 15% GLM: TS-AUC 0.6060 versus 0.6053 for its matched LightGBM control, with both half-bootstrap intervals crossing zero. These small diagnostic gains were not accepted.

The later pairwise-logistic blend is a ranking surrogate. I sample positive/negative pairs only at the same online time, allocating the 100,000 sampled pairs in proportion to $n_+(t)n_-(t)$. For score margin m between paired observations and temperature T=0.05, its loss is:

$$L_{pair} = \text{mean}\{\log[1 + \exp(-m/T)]\} + \text{regularization}. \tag{11}$$

A five-coefficient sigmoid gate maps four training-standardized causal covariates—log online length, online/history ratio, first-order autoregressive (AR1)-adjusted effective length and historical null NLL—to a weight bounded by 0.5. An analytic gradient and the bounded limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm (L-BFGS-B) optimize the surrogate. Fit-only nested student predictions are necessary because fitted-on-the-same-data scores would bias gate learning. The evaluated variant loses on average 0.002429 against its student control and 0.004850 against fixed zone_v5; all gates fail. Its historical GRU features also inherit the earlier provenance limitation.

Exact TS-AUC optimization was subsequently tested separately using differential evolution, first for the original null and later for the nested selected-width gate. The later adaptive gate loses 0.003679 against the fixed-rule selected-width combination. Thus neither a more metric-aligned objective nor an additional calibration learner guarantees better outer-fold ranking. My selection hierarchy is: exploratory fit diagnostics and contribution screens; training-only nested configuration choice; frozen outer comparisons across folds/seeds; then budgeted confirmation and streaming parity. Lower NLL alone cannot skip downstream model selection.

3.9 Model families investigated

My model search covered both classifiers on engineered features and models operating on the observation sequence. Tree methods included Random Forest (RF), LightGBM, categorical boosting (CatBoost) and extreme gradient boosting (XGBoost). Neural networks (NNs) included feedforward multilayer perceptrons (MLPs), feature-based recurrent neural networks (RNNs), raw-sequence convolutional neural networks (CNNs), temporal convolutional networks (TCNs), and pretrained forecasting encoders. The recurrent branch covered a dense GRU, a dendrite-inspired architecture with separate memories for feature groups, and independently recurrent neural network (IndRNN) cells with independent per-neuron recurrent weights. The dendritic model was my recurrent adaptation of sparse branch-to-output wiring, rather than a reproduction of an unchanged feedforward dendritic network. I also investigated simpler logistic models, reservoir computing and kernel/memory-based alternatives. This inventory describes explored families; differing feature banks, data subsets and validation stages prevent a single fair numerical ranking of every historical run.

Model family	Tested role and evidence
LightGBM	Main feature classifier; bank, capacity, objectives, soft labels and ensembles; selected submission 46
Random Forest	Feature-classifier and blend experiments, including the RF256 submission lineage; later matched LightGBM/RF blends lacked statistical confirmation
CatBoost / XGBoost	Alternative boosted trees, hyperparameter screens and blends on overlapping feature banks; no accepted replacement for the selected model
Feedforward NN / MLP	Neural feature classifiers and family screens; extent of completed validation varies by recipe, so no universal neural-versus-tree verdict is claimed
DenseRNN and DendriticRNN	Dense GRU control versus recurrent feature-group branches; an early matched three-fold panel scored 0.5226 and 0.5511 respectively, both local
IndRNN	Independent-neuron recurrent cells in the dendritic architecture, with saved full-fold predictions and separate proxy hyperparameter screens; no accepted final replacement
Raw-sequence CNN / TCN	Causal convolutions learned directly from values; a 30-epoch historical screen reached local validation TS-AUC 0.5294
Feature-based TCN	Causal convolution over engineered features; ensembles, normalization, ranking-loss and regularization variants; public submission 49 remained below 46
GRU forecasting null	Learned normal-process residuals, Optuna, nested width choice, fixed/adaptive blending and sequential evidence; detailed matched results in Sections 5.2 and 6
Foundation models	Chronos/Bolt, the Time Series Foundation Model (TimesFM) including a financial checkpoint, and FinCast; forecasting, probes and limited fine-tuning; scope and negative outcomes in Section 5.4
Logistic GLM / pairwise-logistic gate	Coefficient and interaction diagnostics, probability blends and ranking surrogates; distinct objectives described in Section 3.8
Echo-state network (ESN)	Fixed recurrent reservoir with a trained ridge readout; a 3,000-series historical panel scored pooled OOF 0.5840
Hierarchical temporal memory (HTM)-style / multilayer kernel machine (MKM) methods	Streaming surprise and stacked approximate-kernel alternatives; the historical HTM-style screen was near random ranking, and the MKM blend retained baseline weight 1
Statistical and probabilistic models	AR/BIC, GARCH-family and generalized autoregressive score (GAS) nulls with Student-t innovations (GAS-t), GLRs, CUSUM, Bayesian evidence and oracle-calibrated combinations; used as representations, diagnostics or classifier inputs

Table M1. *Model coverage established from implementation and saved experiment records. Historical diagnostic scores are explicitly local and use their own protocols. An explored family is not evidence that every architecture, optimizer or fine-tuning schedule within it was exhausted.*

3.10 Oracle revisions and what they established

The files named oracle_ceiling contain a sequence of approximate probabilistic models, not a known optimal solution for the unknown DGP. The initial model integrated mean- and variance-shift likelihood ratios over candidate break locations, calibrating population amplitude grids on training folds. Subsequent revisions tested shrinkage of measured amplitudes, maximum rather than additive aggregation across scales, autocorrelation and distribution-shape channels, a training-calibrated break-timing prior, and joint Bayesian online change-point detection (BOCPD)-style marginalization

in which channels agreed on the same candidate location. The later whitened-residual channel reused the BIC-AR/GARCH/PIT representation. These revisions tested specific explanations for the gap to the feature classifier; adding a channel or changing its aggregation was not inherently beneficial.

Historical oracle revision	Local pooled OOF TS-AUC	Finding
Initial mean + variance sum	0.5172	Restricted likelihood model left a large gap to the feature classifier
Shrinkage-adjusted amplitude grids	0.5165	This correction did not improve the combined score
Maximum-over-scale mean + variance	0.5398	Aggregation choice mattered
Added autocorrelation / CDF channels	0.5476 for variance + autocorrelation	Adding all channels together was worse than the selected pair
Learned four-channel logistic combiner	0.5320	A learned combination did not recover the best fixed-pair score
Joint shared-location variant	0.5510 for the reproduced variance + autocorrelation pair	Sharing a candidate location changed the evidence combination
Whitened moment + autocorrelation channels	0.5800	Improved its own 0.5510 oracle control, still below the historical whitened LightGBM reference 0.6053

Table O1. *Saved historical oracle updates, not public leaderboard scores and not mathematical performance ceilings. The last selected combination passed one logged two-half bootstrap comparison against its own oracle control. Selection among combinations used the same OOF results and that gate did not correct for the full selection history; it is not the later five-seed acceptance standard or confirmation against submission 46.*

Two other diagnostics must be kept separate. A time-only prior cannot discriminate among series at a fixed online time under TS-AUC, although a location prior inside joint evidence marginalization can alter the ranking. A variant normalized by a series' eventual total online length used future information and was explicitly labelled a non-causal diagnostic, not an admissible inference feature. Likewise, the later mean-break oracle that knew each evaluated series' true break location was retrospective. None of these constructions supplies the exact DGP or licenses forward-looking predictions. Oracle channels also informed larger feature banks and teacher experiments; the subsequent 731-column nested bank lost to smaller banks, reinforcing that extra approximate oracle information did not automatically improve model selection.

3.11 Blending, stacking and complementary predictions

I investigated three ways to combine models. A probability blend averages their outputs, for example $s = w\,p_1 + (1-w)\,p_2$ with a fixed scalar w. An adaptive blend learns w from causal context and model disagreement. Stacking instead feeds base-model predictions into a second learner; the historical dendritic RNN received the LightGBM prediction as an additional input branch alongside engineered features. Averaging seeds of the same learner, concatenating feature families, and teacher–student soft-label training are related ensemble or representation choices, but are not this prediction-input stacking procedure.

Historical probability sweeps combined LightGBM with RF, CatBoost, XGBoost, a logistic GLM, recurrent networks and kernel/reservoir alternatives. Their saved local screens illustrate the difference between an average gain and confirmed complementarity: a 50/50 CatBoost blend scored

0.6102 against its 0.6053 LightGBM control, and a 50/50 RF blend scored 0.6082 against that same control. Neither passed both half-bootstrap tests; decoupled split/bootstrap retests also rejected these saved blends. The 85/15 GLM blend reached 0.6060 without acceptance. These pooled historical scores belong to their own feature banks and cohorts, rather than the recent submission-46 comparison.

The combined whitening-and-stacking RNN screen used 3,000 series and three group folds. Its recurrent score was 0.5503, versus 0.6067 for LightGBM on the common series; the subsequent blend sweep selected weight 1 on LightGBM, retaining no recurrent contribution. The implementation loaded previously computed LightGBM OOF predictions rather than refitting the base learner inside each recurrent outer-training pool. It also used outer validation during neural epoch evaluation. Accordingly, this historical screen does not establish fully nested, selection-independent stacking performance. OOF predictions exclude their own series during base fitting, but may still depend on the meta-learner's outer validation series unless the base fits are nested within that outer training pool.

Later combination experiments used the trial312 student with a separate residual-feature classifier: fixed 80/20 blending before t=500, training-only late-weight selection, and learned sigmoid gates fitted either to a pairwise-logistic surrogate or exact calibration TS-AUC. The selected-width fixed rule reached mean matched search TS-AUC 0.622489 against 0.619308 for trial312; the exact adaptive rule fell to 0.618810. The historical family-score forward build lost approximately 0.0110 against its submission-46 control (Section 3.7.1). The feature-based TCN ensemble in public submission 49 scored 0.6171, below submission 46's 0.6191. Thus flexible combination did not uniformly improve on a simple fixed rule; recent incremental and fixed-holdout tests remain the basis for judging replacement.

My validation requirement is to generate meta-training predictions by series-level inner cross-fitting within each outer training pool, fit score transforms and combination weights there, then evaluate the frozen combination on the outer validation series. A weight selected on all scored OOF series and tested on those same series is an exploratory sweep. Even a separate weight-selection panel does not repair upstream base-model leakage; independence must hold throughout the hierarchy. The recent nested residual experiments satisfy the recorded group-separation protocol, while their fold-relative percentile transforms still require streaming-export validation (Section 3.2).

3.12 Residual transfer within and across model families

I also asked whether a second learner could explain what a first model missed. Two residual constructions answer different questions. A classifier's training error is $r=y-p^{A}$, whereas a forecasting null produces an observation innovation $e=x-\hat{\mu}^{A}$, optionally divided by its predicted scale. The former requires a known training label; the latter can be computed when the next observation arrives. My classifier-correction experiments trained B to predict r from the original causal features and added its predicted correction to A:

$$r_{i,t} = y_{i,t} - p^{A}_{i,t};\ s_{i,t} = p^{A}_{i,t} + k\,\hat{r}^{B}_{i,t}. \tag{12}$$

The true label residual is a training target, not an online input feature. At inference, only B's predicted correction is available. Forecast-residual experiments instead pass observed innovations and their causal summaries into the downstream break classifier. Both can cross model families, but they are not interchangeable with prediction-input stacking or ordinary averaging.

Route	Information transferred to the next learner	Recorded outcome
LightGBM classifier → RF regressor	OOF y−p residual as B's training target; the original 84 features remain its inputs	k=1 scored 0.5995 versus 0.6053; the sweep selected k=0, with no RF correction retained
Small LightGBM → larger LightGBM regressor	Same-family residual target; seven-leaf first stage versus 127-leaf correction stage	The sweep again selected k=0; 0.6074 versus the historical 0.6053 control reflected the first stage, not a successful residual correction; gate rejected
AR/GARCH statistical null → feature classifier	Mean/scale-adjusted observations, transformed residual statistics and change evidence	Residual representations form part of the selected feature bank; their contribution is context-dependent in the historical ablations
GRU forecasting null → LightGBM classifier	Standardized innovations, likelihood and mean/scale/dependence evidence; later multiscale and sequential summaries	The selected-width combination reached 0.622489 on matched search folds; later sequential evidence reached 0.623224 without incremental acceptance
Null-feature LightGBM → student LightGBM	Null logit supplied as init_score, then fitted trees predict an additive logit correction	Lost roughly 0.016–0.018 in the tested historical recipe; the separate blend was preferable
LightGBM → dendritic RNN	A's probability supplied as an input branch, rather than y−p	Prediction-input stacking, described in Section 3.11; the final historical blend retained zero recurrent weight

Table R1. *Verified transfers within one learner family and across different architectures. The routes include different residual definitions, feature banks and evaluation stages; this is an experiment inventory rather than one homogeneous score ranking. I did not establish every possible A→B pairing as tested. Classifier-correction records are src/cv_score_rf_residual_bic_ar_rosenblatt.py and src/cv_score_lgbm_two_stage_bic_ar_rosenblatt.py, with their matching saved logs.*

The classifier-residual scripts reused global OOF predictions for A and the same series folds for B. Although each row's A prediction excluded that row's series, residual targets for B's training series could still depend on B's outer validation series through other base-model fits. The two-stage LightGBM also used outer validation for early stopping, and both scripts selected correction strength on the same pooled results later gated. These are historical exploratory screens, not fully nested confirmation. A strict successor would construct A's training predictions entirely within B's outer training pool, select k there, and freeze both stages before outer evaluation. The newer nested GRU route addresses shared-model group separation separately; streaming score transforms and fixed-holdout confirmation remain distinct requirements.

4 Experimental design and statistical analysis

Recent comparisons use five outer series-level folds and tree seeds 6, 7 and 8 on the fixed search universe. Each reported search score is a mean of fold TS-AUCs, subsequently averaged across tree seeds. Trials and feature caches are frozen before evaluation; multiple related methods are not merged after observing their outer-fold results.

For acceptance, I use paired whole-series cluster bootstrap, recomputing the actual TS-AUC after resampling. A project-specific two-half rule requires both corrected confidence intervals for candidate-minus-control to lie above zero. Five bootstrap seeds assess sensitivity to the

partition/resampling realization; they do not constitute five independent studies. The general bootstrap principle follows Efron [9]; the two-half acceptance rule is my own conservative operational criterion.

Each later comparison uses 300 resamples per bootstrap seed. Bonferroni correction counts differ by campaign: 12 for the early/late/multiscale tests, 16 for the adaptive gate and 20 for sequential residuals. Therefore gate counts across these campaigns are not directly comparable. These corrections address the declared comparison family, not all accumulated researcher selection.

The fixed 2,000-series holdout has a twenty-evaluation project budget, of which five confirmations are registered. I also assess the previously used reduced panel and five preregistered synthetic process families. Neither is presented as independent real-domain validation. Width and trial312 were selected on the search universe; changing folds and seeds does not remove that historical selection bias.

4.1 Validation and implementation audits

Two major corrections affected earlier local conclusions. First, a reduced scorer used absolute rather than relative online time: earlier RF sanity scores near 0.66 became 0.52054 and 0.52167 after correction. Public scores were unaffected. Second, a teacher/student evaluation reused shared folds. At $\alpha=0.5$ its score fell from 0.64721 to 0.61743 under nested teacher cross-fitting; at $\alpha=0.85$ it fell from 0.65423 to 0.61120. Older learned-null caches also required stricter separation of holdout labels from online normal-material censoring.

I check label offsets, group separation, row alignment, reset behavior, prefix invariance and finite outputs. Earlier union152 streaming/batch checks found maximum differences near 1.1×10^{-13}. For sequential evidence, future-value perturbations leave prefix features unchanged; long extreme streams remain finite; neutral log-evidence satisfies $R_t=t+1$. Strict graphics processing unit (GPU) checkpoint loading and real feature/classifier smoke tests precede full runs.

4.2 Hyperparameters, search algorithms and GPU likelihood evaluation

Hyperparameterization operates at several levels: per-series null-model likelihood parameters; feature banks and residual transforms; learner capacity and regularization; training-material/window choices; and blend weights. Their objectives differ. GAS-t fitting minimizes history NLL; GRU Optuna initially minimizes validation NLL; LightGBM/soft-label and later width/blend searches target grouped downstream TS-AUC. I record the objective and split for each search rather than treating all optimizer winners as detection improvements.

Search	Documented configuration or space	Objective and outcome
LightGBM campaign	Learning rate 0.015–0.12 log scale; leaves 7–127; minimum leaf rows 30–2,000; feature fraction 0.4–1; bagging 0.5–1; L2 0.1–50; extra trees on/off	Four grouped search folds; rounds scale inversely with learning rate
Soft-label campaign	α in [0,0.95], leaves 7–127, extra trees on/off	Selected trial312: α=0.14595873, ten leaves, extra trees; later matched validation reported separately
Experiment-design controller	Multivariate tree-structured Parzen estimator (TPE), eight startup trials; Hyperband over one to four folds; cost-aware Thompson sampling across campaigns	489 total ledger trials across campaigns; pruning and adaptive allocation are search, not confirmations
Original GRU Optuna	Hidden {32,64,128}, layers 1–3, window {128,256,512}, v 3–15, learning rate 0.0005–0.005, steps 1,000–3,000; normal material and length weighting toggles	TPE seed42, eight startup trials, median pruning; 23 complete and seven pruned; trial26 NLL 1.283579
Nested GRU width search	Hidden {64,128,256}; other v5 knobs fixed	Downstream calibration TS-AUC; fold choices 128/64/128/256/64
Exact adaptive gate	Differential evolution (DE), 30 iterations, population multiplier five, bounded coefficients, fixed regularization/fallback	Training-only exact TS-AUC; outer loss despite calibration search

Table S1. *Representative saved search specifications. Pruned trials do not have complete cross-validation scores. Trial312's selected four-fold search score, 0.622774 versus campaign baseline 0.608102, is not the later matched five-fold score or a new holdout result. Figure 5 plots optimization traces without treating them as learning curves on unseen data.*

4.2.1 LightGBM options beyond the automated search

Table S1 summarizes one automated campaign, not the entire LightGBM investigation. Earlier searches used different banks and ranges: learning rate 0.01–0.15, 15–255 leaves, 50–1,000 minimum leaf rows, row/column fractions 0.5–1, bagging frequency 1–5, L1 leaf regularization 0–10, L2 regularization 0.1–20, split-gain floor 0–1, and maximum depth 4–16 or unlimited. A smaller proxy grid crossed leaves {7,15,31,63} with minimum leaf rows {200,500,1000,2000}. These are separate search specifications, not one jointly evaluated space. Parameter meanings follow the official LightGBM documentation; Table S2 records the broader tested choices.

Choice family	Options investigated	Evidence and scope
Capacity and fitting schedule	num_leaves, min_data_in_leaf, max_depth, boosting rounds, learning rate; fixed-round curves and early stopping	Proxy grids, grouped follow-up fits and the seed/capacity diagnostics in Figure 4
Row and column sampling	feature_fraction, bagging_fraction, bagging_freq; separate seed controls	Historical full-column access helped one bank; row and column subsampling did not behave identically
Leaf and split regularization	lambda_l1, lambda_l2, min_gain_to_split, min_sum_hessian_in_leaf, path_smooth	Targeted default-option sweeps supplemented the automated parameter search
Split resolution and leaf form	max_bin=1023; linear_tree with linear rather than constant leaves	Both lost on the historical full-column union152 reduced panel
Boosting and randomization	Gradient boosting decision trees (GBDT); Dropouts meet Multiple Additive Regression Trees (DART); randomized thresholds through extra_trees	DART/dropout proxy configurations and saved grouped predictions; extra_trees results depended on the bank and target recipe
Training objective and target	Binary log loss; cross_entropy for nested soft targets; time-grouped LambdaRank with binary or delay-graded relevance	Soft-label training underlies submission 46; ranking variants did not establish a replacement
Class and row weights	is_unbalance; late-positive delay weighting; symmetric weights before/after the labelled break	Different training interventions; symmetric weighting gave mean +0.00009 against submission 46 without acceptance
Target ramps and direction constraints	Post-break ramps of 25/50/100 steps; pre-break ramps of 25/50; monotone_constraints from training-fold correlations or RF-derived directions	Tiny or negative local changes; no accepted addition to the selected student
Model combination and augmentation	Two-stage classifier; null init_score; seed averaging; external-model blends; simulated breaks/history shifts plus smaller randomized trees	Separate fixed controls and validation limits; Sections 3.11 and Appendix A summarize outcomes

Table S2. *Verified LightGBM option coverage, including changes to the training problem and surrounding pipeline. These rows do not assert that every LightGBM option or every combination was tested. A source file establishes implementation; completed scores and gates establish the recorded evaluation stage. Ensemble weights and augmentation are not native tree hyperparameters.*

The historical full-column union152 reduced-panel control scored 0.65281, versus 0.63231 with feature_fraction=0.8. Against that 0.65281 control, max_bin=1023 scored 0.63106, path_smooth=10 scored 0.62104, class rebalancing scored 0.62575, DART scored 0.59039 and linear leaves scored 0.57703. Lowering the minimum leaf Hessian sum to 10^{-6} left predictions bit-identical. These are reused local-panel comparisons, not public scores or five-seed bootstrap acceptance. The name gate_sweep_summary in their records denotes a scoring sweep and must not be read as proof of statistical acceptance. In particular, extra_trees lost in this hard-label bank but was retained in the later trial312 recipe; a parameter verdict cannot be transferred unchanged across representations and targets.

On a different hard-label union144 five-fold screen, learning rate 0.02 with 1,500 rounds improved pooled TS-AUC from 0.61166 to 0.61402, winning all five folds but passing 0/5 gates. Adding it to its matched soft-label student changed 0.61858 to 0.61863, only +0.00005. The post-break ramp-25 gave +0.00009; ramp-50/100 and train-correlation monotonicity lost. The earlier RF-sign constraint

screen likewise lost, 0.5764 versus 0.5824. Time-grouped graded LambdaRank variants scored 0.5622 and 0.5636 against their historical binary control 0.6053. Grouping by time aligns which observations are ranked, but does not make that training loss identical to TS-AUC.

Completeness of fits and completeness of acceptance testing are separate. The ten-configuration regularization_student run saved all five-fold predictions, but its retained log records completed gate summaries only through rounds 350 and leaves 31/15. Those three summaries show 0/5 acceptance; unlogged gates are not assigned a verdict. Combined augmentation/regularization recipes were subsequently tested across seeds, yet their early gains did not consistently add to soft-label regularization. Figure 4 therefore presents these records as diagnostics, while Table 2 and the later matched controls establish what actually reached submission or further validation.

4.2.2 Per-series likelihood fitting and population search

The GAS-t implementation follows the score-driven framework of Creal, Koopman and Lucas [17]. In my AR1 mean and Student-t scale parameterization, λ is the log squared scale and the scaled likelihood score drives its update:

$$u_t^2 = \varepsilon_t^2 \exp(-\lambda_t);\ b_t = (\nu+1)u_t^2/(\nu+u_t^2) - 1. \tag{13}$$

$$\lambda_{t+1} = \omega + \beta\lambda_t + \alpha b_t;\ \varepsilon_t = x_t - (c + \varphi x_{t-1}). \tag{14}$$

Here b is bounded between −1 and ν. The Student-t squared scale exp(λ) is not its variance: for ν>2 the variance also contains ν/(ν−2). The existing module sometimes calls it log variance; this report states the likelihood parameterization explicitly. GAS is the model family, not a global-search algorithm.

An early twenty-history screen recorded several optimizers, but it cannot rank them. The global methods used a box of ±5 on raw persistence, which caps β at about 0.9923, while unconstrained Nelder–Mead could approach 0.999. Near-zero score coefficients were also less accessible inside that box. The large negative NLL changes in Table S3 are therefore the signature of a narrower feasible set, not a demonstration that differential evolution, dual annealing or TPE cannot fit GAS-t. Nonzero fitted coefficients were sometimes described as “fixing degeneracy”; that is not evidence of a better likelihood. Multi-start Nelder–Mead was kept as a practical local recipe, not as a certified global optimum.

Historical optimizer, as run	Family	Mean seconds/history	Mean NLL change vs single-start
Single-start Nelder–Mead	Local simplex, unconstrained persistence	2.68	Reference
Multi-start Nelder–Mead	Local restarts, same unconstrained domain	14.77	+1.109
Differential evolution	Evolutionary search, persistence boxed at ±5	46.46	−99.461
Dual annealing	Generalized simulated annealing, same box	46.62	−99.462
Optuna TPE, 150 proposals	Surrogate search, same box	0.91	−171.525

Table S3. *Diagnostic of an unequal-domain screen, not a ranking of optimizer quality. Positive values mean lower NLL. Later equal-bound comparisons are the GPU pilot (Appendix D) and the 8,000-series downstream fit (Section 5.5).*

Differential evolution mutates/recombines a population [18]; particle swarm optimization (PSO) moves particles toward their own and the population's best locations [19]; dual annealing combines stochastic exploration with local search [20]; TPE uses a surrogate model of promising parameter regions [10]. DE is evolutionary search and is distinct from particle swarm; the older script's broad "swarm-family" wording is imprecise. Covariance matrix adaptation evolution strategy (CMA-ES) and genetic algorithms are possible alternatives, but no completed equal-domain runs for them are established by these artifacts.

Comparable optimizer evidence begins only when the search domain is shared. I therefore ran a bounded GAS-t GPU pilot. The full-history recurrence runs in a 64-bit floating-point (FP64) kernel using the Heterogeneous-computing Interface for Portability (HIP), parallel over 128 parameter candidates while remaining sequential within each history. Eight fixed search-only histories are evaluated with seeds 6/7/8, 100 generations of DE/rand/1/bin (mutation factor F=0.7, crossover probability 0.8) and particle swarm (inertia 0.7298, cognitive/social 1.49618), plus bounded local restart controls and L-BFGS-B polishing. Both population methods receive the same local anchor; their reported run time excludes the common central processing unit (CPU) anchor cost. All selected NLLs are re-evaluated with the original CPU objective. Expanded raw parameter bounds are identical across these new methods, and no labels or online observations enter this fit. Figure 6 and Appendix D report the completed eight-history pilot. A finite population search does not certify the global optimum. I subsequently tested downstream residual features on all 8,000 search series, with the separate design and results below.

The downstream experiment fits every series using its own history only, without shared fitted parameters or label-dependent censoring. Four bounded Nelder–Mead starts, their local polish, and DE plus polish share the same bounds. Unlike the eight-history pilot, all methods additionally require $\beta-\alpha(\nu+1)/4>-0.99$: this bounds the lower derivative of the log-scale recursion, while its upper derivative is $\beta<0.999$. The restriction was introduced after an extreme-parameter smoke test exposed numerical sensitivity. Hence the full experiment evaluates a modified stable domain, not a simple enlargement of the pilot cohort. The AR(1) coefficient remains restricted to $[-0.5,0.5]$.

A single fixed optimization seed 6 and 128 candidates over 100 generations determine the per-history DE fits. A continuous fitted-initial-scale recursion processes history and the online prefix; the last 500 historical residuals calibrate mean/scale. Eleven causal change-evidence and likelihood

summaries feed five-fold LightGBM classifiers with tree seeds 6/7/8 and 600 rounds. The frozen combination uses 80% trial 312 and 20% GAS-derived score before t=500, then trial 312 alone. Controls reproduce the same IDs, times, targets and folds. Seven declared comparisons per tree seed, five bootstrap seeds and 300 resamples yield 105 gates, with two-sided corrected intervals at $\alpha=0.05/7$. This does not correct for all historical research reuse or constitute three independent optimizer realizations.

Tree size, seed averaging and burn-in also require context. Figure 4 reports a historical round curve, same-seed tree-capacity comparisons, individually measured seeds versus their probability average, and online ratchet burn-in. Online burn-in delays score smoothing; AR burn-in makes order likelihoods comparable; optimizer startup and neural warm-up are separate concepts. Increasing rounds or averaging predictions is not selecting the best validation seed. Early stopping on sparse cross-validation (CV), reduced-panel rankings and public scores must remain separate: their misalignment motivated nested controls and exported-package checks.

5 Results

5.1 Recorded public submission results

Submission	Model change	Recorded public score
31	RF256, external84 features	0.5974
37	Configuration selected on reused local research scores	0.5107
40	LightGBM union152	0.5984
41	Feature sampling, rounds and smoothing changes	0.6099
42	Context columns removed	0.6129
45	Trainable package, five seeds	0.6164
46	**Nested soft-label student**	**0.6191**
49	Student/TCN ensemble with additional regularization	0.6171

Table 2. *Selected points in the submission lineage. Submissions 47 and 48 were uploaded but never scored. The table is historical public evidence; private final evaluation is not claimed here. The local advantage of the more flexible TCN package did not transfer over submission 46. The run window closed in early October 2026 and the selected recorded public score remains 0.6191.*

5.2 Matched offline comparisons

Method	Mean search TS-AUC	Increment vs selected-width combination
Submission 46 control	0.614015	—
Trial312 student	0.619308	—
Student + nested fixed64 null	0.620524	—
Student + selected-width null	0.622489	Reference
Early null off	0.622696	+0.000207
Late null on	0.622263	−0.000227
Multiscale residuals	0.622899	+0.000410
Adaptive confidence gate	0.618810	−0.003679
Sequential residuals	**0.623224**	**+0.000734**

Table 3. *Matched recent search comparisons. No listed extension passed all incremental acceptance tests. An increase of 0.000734 AUC is 0.0734 percentage points, not a 0.734-point improvement.*

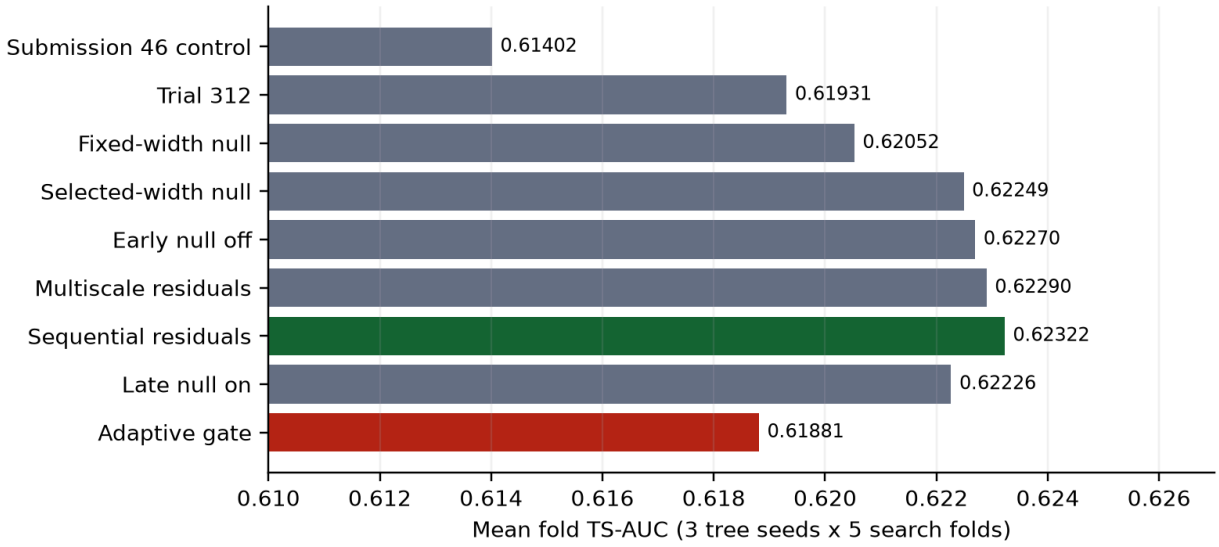


Figure 1. *Average fold TS-AUC on the same search cohort. The axis is truncated to show small differences. Bars are descriptive means, without confidence intervals; they do not represent independent confirmations.*

Sequential residual gains over the selected-width reference are +0.000934, +0.000788 and +0.000481 for seeds 6, 7 and 8. Each wins three of five folds. Against multiscale, gains are +0.000411, +0.000520 and +0.000043. Every incremental gate is 0/5 for every tree seed against either comparator. Against the submission 46 control, sequential residual gate counts are 0/5, 3/5 and 0/5 under the twenty-comparison correction.

For tree seed 6 and bootstrap seed 0, sequential evidence versus the selected-width reference has half-sample deltas +0.001275 and +0.000403. The corrected intervals are [−0.002181, +0.004700] on 3,999 series and [−0.003543, +0.004198] on 4,001 series. Both include zero, so this gate fails

despite positive point estimates. These intervals describe the half-sample TS-AUC estimator, not the mean-fold table. With 300 resamples and strict corrected tail probabilities, finite Monte Carlo resolution is itself a limitation of the acceptance estimates.

The adaptive gate improves its inner calibration objective in three folds but loses in all fifteen outer seed/fold cells. This discrepancy is consistent with unstable transfer of a flexible calibration-selected rule; the experiment does not isolate its cause.

5.3 Fixed holdout and stress panels

Configuration	Rounded fixed-holdout TS-AUC	Rounded delta vs submission 46	Gates
Trial26 zone	0.615	+0.004	0/5
Trial312	0.614	+0.003	0/5
Clean zone_v5	0.616	+0.005	0/5
Trial312 + clean zone_v5	0.620	+0.009	0/5

Table 4. *Fixed confirmations registered once per configuration. Scores and deltas were stored rounded and need not subtract exactly after rounding. The selected-width, temporal and sequential extensions have not been confirmed on this holdout.*

A mixed 10K diagnostic combines 8K OOF with saved 2K holdout predictions and gives 0.621374 versus 0.613142 for its matched control. Different fits and cohort-specific ranks make this descriptive rather than an independent 10K confirmation. On the previously consulted 100-series reduced panel, the clean combination gives 0.56049 versus 0.55352; clean zone alone gives 0.56590. On 100 synthetic series spanning AR, independent and identically distributed (iid), seasonal, Student-t and trend families, the clean combination gives 0.77092 versus 0.73108. Synthetic gains do not establish transfer to new real-world domains.

5.4 Forecasting and foundation-model experiments

Optuna [10] optimized normal-history GRU NLL, with trial26 reaching 1.283579. Its downstream blend gains +0.002879 against submission 46 but only +0.000458 against the older zone_v5 control, with 0/5 gates. The comparison also changes training-material provenance, so it is not a hyperparameters-only ablation.

On a common 1,000-series panel, rolling forecast-null TS-AUCs are 0.504 for Bolt P16, 0.508 for Bolt P4, 0.502 for FinCast and 0.493 for TimesFM, versus 0.543 for the GRU null. Forecasting references [11–14] motivate these models, but their forecasting results do not imply break-detection performance.

Supervised representation	Mean blend delta vs submission 46	Delta vs nested fixed64 combination
Financial TimesFM, frozen probe, 3 seeds	−0.003254	−0.004632
Financial TimesFM, limited finetune, 3 seeds	−0.002697	−0.004283
FinCast, frozen probe, 3 seeds	−0.000675	−0.001693
FinCast, limited finetune, seed 6	−0.003770	−0.005329

Table 5. *Task-specific classifier heads and limited encoder adaptation. FinCast finetune seeds 7/8 were intentionally dropped for cost; no results are attributed to them. Financial TimesFM uses pfnet/timesfm-1.0-200m-fin [13]; strict converted-checkpoint loading and query/key/value (QKV) projection checks passed, but complete forward-output parity between the JAX automatic-differentiation library and PyTorch was not established.*

Independent limited Bolt finetunes produced mean +0.001199 against submission 46 and −0.001222 against the older zone control, with all gates 0/5. Their small benefit remained unaccepted. Chronos-2 was explored and dropped before a full benchmark. These outcomes apply to the tested recipes, not all possible time-series foundation models or adaptation budgets.

5.5 Full-search GAS-t optimization and downstream detection

The 8,000-series run completed on 13 October 2026 at 22:00; its independent post-run audit and completed-result code review passed on 14 October. Every per-history fit excludes online observations and labels, and the fixed 2,000-series research holdout was excluded throughout. Inputs, frozen sources and holdout-ledger hashes match. An additional review independently reproduced all 135 saved fold-score values using within-time scikit-learn AUCs, checked all 8,000 stored parameter sets for finiteness/stability, verified exact control predictions and blend cutoffs, and checked all 105 gate records. These checks establish numerical/data consistency, not independent confirmation of a model improvement.

Matched method	Mean fold/seed TS-AUC	DE blend minus this row	DE acceptance vs row
Submission 46 control	0.614015	+0.006661	0/5 for every tree seed
Trial 312 student	0.619308	+0.001368	0/5 for every tree seed
Student + local GAS fit	0.619315	+0.001361	0/5 for every tree seed
Student + locally polished GAS fit	0.619810	+0.000866	0/5 for every tree seed
Student + DE/polished GAS fit	0.620676	—	Reference
Student + selected-width GRU null	0.622489	-0.001813	0/5 for every tree seed
Student + sequential GRU evidence	0.623224	-0.002548	0/5 for every tree seed

Table 6. *Matched full-search detection comparisons. Values are the mean of five fold scores and three tree seeds, with identical cached controls. Seven comparisons per seed include a separate DE-null versus locally polished null comparison. Every one of the 105 gates failed; no holdout confirmation, public score or promotion is claimed.*

DE plus polish improves its matched blend by 0.000866 over local polish alone and wins 9/15 fold–seed cells. Against trial 312 the mean gain is 0.001368 with 11/15 wins; against submission 46 it is 0.006661 with 12/15 wins. However, it trails the selected-width GRU combination by 0.001813 and sequential GRU evidence by 0.002548, winning only 6/15 and 3/15 cells respectively. The standalone null classifier scores 0.562205 versus 0.559395 under local polish, but that comparison also fails every gate. A better raw likelihood does not determine which residual representation best complements the student.

On the same 8,000 histories, DE lowers summed whole-history NLL by a mean 7.226998 relative to unpolished local fits and 3.198186 relative to local polish. At tolerance 10^{-6} it beats local polish on 3,521 histories and loses on 1,849; the remaining 2,630 are within tolerance. These are sums over each complete history, not per-observation forecasting losses, and cannot be compared numerically with the GRU's 1.283579 validation objective. The population includes the unpolished local anchor, making non-regression against that anchor a protocol property rather than proof of global optimality. Local polish was not itself injected as the DE anchor, so it can still beat the finite population search.

For DE blend versus local-polish blend, tree seed 6/bootstrap seed 0 gives half-sample deltas +0.000357 and +0.001303 with corrected intervals [−0.003414, +0.004348] and [−0.001431, +0.004126]. Both include zero. The 105 failures therefore preserve the distinction between a positive average and accepted evidence; failure is not proof of zero benefit. These resamples condition on the saved predictions, rather than refitting the full model hierarchy. As in earlier campaigns, 300 resamples provide limited resolution at the corrected tails.

The controller initially misclassified completion because a PowerShell process refresh returned a null exit code, skipping the automatic audit. The frozen computation and all three seed result files were complete. I performed the audit manually and corrected the controller's process-handle handling without altering scientific sources or rerunning the experiment. The source/input provenance and recovery records are retained separately. The frozen implementation, audit, predictions and code review are retained in src/gas_t_downstream_validation.py and artifacts/gas_t_downstream_validation/.

6 Break analysis

I stratify by time, history length, retrospective break type/strength and age. Type and age analyses retain all common negative controls; subgroup AUCs do not sum to the aggregate score. The selected-width combination scores 0.52757 on mean-dominated breaks, 0.56256 on weak breaks and 0.53649 within the first 25 post-break observations.

Online phase	Selected-width TS-AUC	Sequential residual increment
t<100	0.54814	+0.00515
100–249	0.61217	−0.00121
250–499	0.63217	+0.00137
t≥500	0.65464	0

Table 7. *Sequential accumulation improves the early bucket, but regresses in the first middle bucket. The late difference is identically zero by the blend cutoff, not evidence that residuals could never help late.*

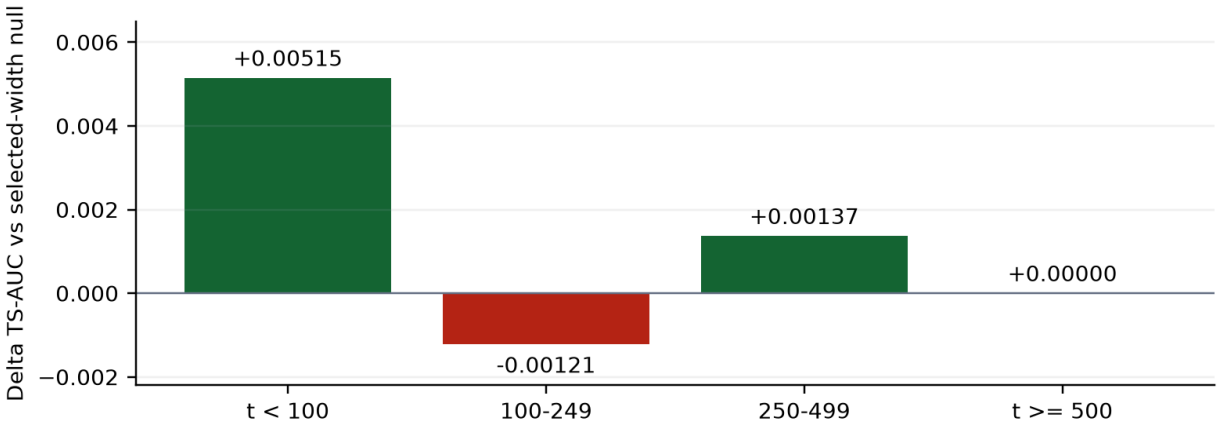


Figure 2. *Mean incremental TS-AUC by phase, averaged over fifteen fold/seed cells. These reused search-data subgroup findings are exploratory and were not used to retune the evaluated recipe.*

Detection diagnostics use the top 20% scores within each fold/time cross-section. Among break series with sufficient scored follow-up, detection within 25 observations is 20.17% for selected-width null, 20.51% for multiscale and 20.59% for sequential residuals. I also save within-50/100 detection, pre-break alarms and never-break ever-alarm rates. These are relative-rank diagnostics, not deployed thresholds or calibrated false-alarm probabilities. Delays are quantized by step25 sampling, and ever-alarm rates depend on exposure.

A mean-break oracle using the true change location scored about 0.663 on a retrospective panel, while existing mean statistics were already near 0.67. This is a comparison with that oracle construction, not a mathematical upper bound. A nested mean-specialist blend was subsequently rejected, supporting caution about another mean-only detector.

6.1 GAS-t: early improvement and middle-phase regressions

The GAS result is heterogeneous across online phases. The DE blend improves on its local-polish control in all three pre-cutoff buckets, yet only the earliest bucket beats the stronger GRU combinations. This helps explain why better history fitting yields a weaker aggregate detector than the existing sequential GRU route.

Online phase	DE blend TS-AUC	Delta vs local polish	Delta vs selected GRU	Delta vs sequential GRU
t<100	0.554483	+0.001988	+0.006339	+0.001193
100–249	0.607279	+0.001290	-0.004893	-0.003684
250–499	0.629681	+0.000738	-0.002492	-0.003861
t≥500	0.654641	+0.000000	+0.000000	+0.000000
Post-break age 0–24	0.537737	+0.000786	+0.001243	-0.001626

Table 8. *Exploratory bucket means over the same 15 fold–seed cells. Fresh-break scoring retains all negative controls at comparable times. Scores within these subgroups do not add up to the aggregate metric, and no separate subgroup acceptance campaign is claimed. Zero late differences are imposed by the common t=500 blend cutoff.*

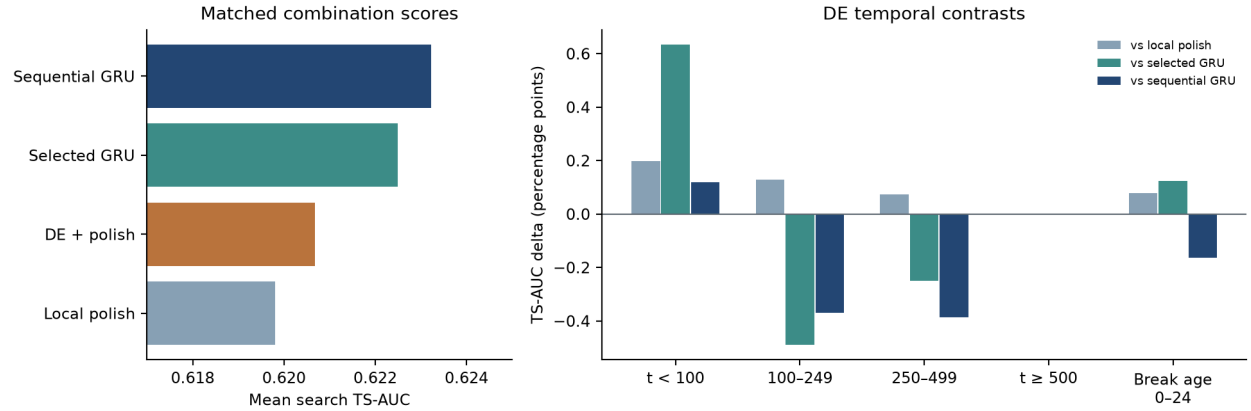


Figure 7. *Left: matched overall combination scores, with a truncated axis. Right: phase/fresh-break deltas against local-polish, selected-width GRU and sequential GRU controls, in TS-AUC percentage points. Means are descriptive; bars have no confidence intervals and reuse the search data. The early advantage is a hypothesis for a separately frozen validation experiment, not permission to choose an early-only weight after inspecting these results.*

The DE blend's early gain is 0.006339 against selected GRU and 0.001193 against sequential GRU, but its middle-phase deltas versus sequential GRU are −0.003684 and −0.003861. For break age 0–24 it beats selected GRU by 0.001243 and loses to sequential GRU by 0.001626. Improving one phase therefore does not establish superior rapid detection overall. These retrospective findings were not used to retune the evaluated combination.

7 Discussion and limitations

The completed GAS experiment reinforces the separation between fit and task performance: broader history-likelihood search improved its own residual blend but did not beat the existing GRU combinations or pass acceptance. Its early advantage and middle-phase loss suggest context-specific complementarity; this was not independently validated or used to retune the frozen rule. Table S3 still cannot support a ranking of global search algorithms, because those historical runs used a tighter persistence box than Nelder–Mead. The later DE comparison is only within a shared, stability-restricted domain, with one optimizer seed, a fixed temporal cutoff and fold-relative ranks. Those limits, together with the restricted AR1/Student-t null, prevent a general verdict about GAS or about global optimization.

The strongest practical contribution of the submitted model was the combination of complementary representations and nested soft-label regularization. Separate learned residuals subsequently added useful search signal. Sequential accumulation yielded the largest recent average search score and addressed part of the early deficit, but did not achieve broad fold consistency or incremental statistical acceptance.

Three limits are central. First, extensive reuse of the search universe creates selection bias that new seeds and stricter cross-fitting cannot remove. Second, sparse local scoring and fold-relative percentile blends are not complete streaming-export validation. Third, the newest representations lack fixed-holdout confirmation and independent domain tests. The rounded older holdout gains cannot be assigned to the later residual model.

The acceptance rule is conservative. Failure means that the chosen procedure did not establish a positive effect in both halves; it is not proof of zero benefit. At the same time, lowering the threshold after seeing failures would weaken the evidential standard. I therefore preserve both the numerical improvements and the non-acceptance verdict.

After 14 October 2026 I ran no new detector comparison on the ADIA search folds and used no further fixed-holdout evaluation. Five of the twenty holdout slots remain registered.

Compared with previous-edition solution reports [3], the main methodological difference is availability of data at prediction time. Complete post-boundary segment comparisons and their ROC-AUC cannot be transferred directly to unknown-location causal monitoring. A 2025 winning score near 0.90 is not a target against which my 0.6191 TS-AUC submission should be ranked.

8 Conclusion

The selected submission achieved recorded public TS-AUC 0.6191 and remains the selected result as of 28 October 2026. Offline development increased a matched local control from 0.614015 to 0.623224 using a student, nested learned-null model and sequential residual evidence. The last extension improves early ranking but loses some middle-phase performance and fails incremental acceptance gates. The separate full 8K GAS-t likelihood-search test yields 0.620676, below the GRU routes, with all 105 gates failing. Improved likelihood alone has not produced an accepted successor. These results identify a promising representation and clarify the validation requirements for a successor; they do not establish a production replacement or improved private leaderboard result.

Research tools

Cursor supported the development workflow; Codex and Claude assisted with implementation, experiment execution, audits and analysis in the shared repository. They are research tools, not independent statistical replications. Detailed experiment-level attribution to early sessions remains incomplete. The saved experiment records remain the source of the reported scores.

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Appendix A Additional experiments and negative results

Experiment family	Observed result	Interpretation
Simulated strong breaks	Approximately +0.0020 in an early multi-seed test; gates 0/5	Unaccepted augmentation signal
History-shift augmentation	Initial gain; corrected teacher variants failed	Earlier t>999 training material was a confound
Regularizer stack	Improved plain baseline, not consistently soft-label control	Gains did not add independently
10% / 30% row subsampling	Approximately −0.014 / −0.009 on sparse rows	Compute reduction cost signal
Symmetric near-break row weights	Mean +0.00009 vs submission 46	No accepted added value
Null as LightGBM init_score	Loss approximately 0.016–0.018	Separate blend preferable in tested setup
Learned alternative adaptation	Mixed small gains, gates 0/5	Complementarity unconfirmed
Allan / PIT-Allan / velocity / Kalman	No accepted added improvement	Standalone usefulness did not guarantee additivity
Family-score forward build	Mean −0.0110 vs its submission-46 control	Compact summaries lost; upstream cache and inner-selection limits remain
CatBoost and larger oracle banks	No accepted replacement in tested comparisons	More flexibility or columns not sufficient
Causal TCN ensemble [15]	Submission 49 scored 0.6171 versus 0.6191 for submission 46	Local advantage did not transfer to public evaluation
Mean-specialist blend	Loss on every outer fold per seed	Mean-only addition was not beneficial overall
Float-grid scan	Own definition found only 25 strong lattice histories	Reported external artifact gain not reproduced
Neural processing unit (NPU) encoder port	Export blocked on unsupported nanmean	No acceleration or model-gain claim

Table A1. *Representative tested families, including negative results. Approximate early deltas belong to their historical controls and should not be compared as one homogeneous ablation matrix.*

The received Discord messages motivated additional hypotheses, including learned nulls and grid-based likelihood accumulation. A reported external grid-ensemble gain was not replicated with my lattice definition. No float-grid signal is included in the residual method. Standalone BIC-AR mean-shift SR features had already been explored; the new test instead uses learned GRU residual mean, energy and lag-product evidence.

Appendix B Reproducibility records

Record	Repository location
Metric and acceptance implementation	src/metric.py; src/acceptance_gate.py
Feature-family loader and selection	src/feature_selection_framework.py
Fixed-holdout ledger	artifacts/expdesign/holdout_log.json
Trial312 matched control	artifacts/expdesign312_vs46/
Clean nested fixed64 null	artifacts/combo_nested_validation/
Confirmations and stress panels	artifacts/validation_20261005/report.txt
Independent Bolt validation	artifacts/bolt_multiseed_validation/
Financial TimesFM and FinCast	artifacts/fm_fin_task_supervised/
Width-selection predictions and choices	artifacts/gru_width_tsauc/
Temporal experiments and adaptive gate	artifacts/gru_temporal_followup/; artifacts/gru_adaptive_followup/
Sequential method, specification and predictions	src/gru_residual_accumulation.py; artifacts/residual_accumulation_followup/
LightGBM option sweeps and targeted tests	lightgbm_historical_options_source.csv; src/regularization_student.py; src/label_monotone_lr_test.py; artifacts/lambdarank_v2_run.log
Ensemble screens and historical stacking	src/blend_lgbm_rnn_stacked_bic_ar_rosenblatt.py; src/train_dendritic_stacked_whitened_cv.py; artifacts/blend_lgbm_catboost_bic_ar_rosenblatt.log; artifacts/retest_borderline_blends_decoupled_gate.log
Normality, distribution and GLM diagnostics	src/moments.py; src/distribution_tests.py; src/glm_diagnostic.py; src/smooth_null_blend_test.py
Historical GAS-t optimizer comparison	artifacts/gas_t_optimizer_comparison_20series.csv; src/test_gas_t_global_opt2.py
GPU GAS-t search, parity and polish-only audit	src/gas_t_gpu_global_search.py; artifacts/gas_t_gpu_global_search/
Figure sources and paper regeneration	artifacts/adia_paper_20261013/build_revision.py; figure3_source.csv; figure4_*_source.csv; figure5_width_source.csv
Recomputed score provenance	artifacts/adia_research_article_20261013/score_provenance.csv

GPU training/inference used a Radeon 890M from Advanced Micro Devices (AMD), using its Radeon Open Compute (ROCm) platform on Windows; LightGBM and bootstrap ran on CPU. Jobs were serialized and cached. GPU launch failures were recovered with batch size four; finite-output repair and an evaluation-import fix did not constitute new scientific configurations. Source and holdout-ledger hashes, per-fold predictions and bootstrap progress support reconstruction of the recent campaign. This is a technical competition report, with no claimed peer review, public code release or complete independent replication.

Appendix C Feature-selection and machine-learning diagnostics

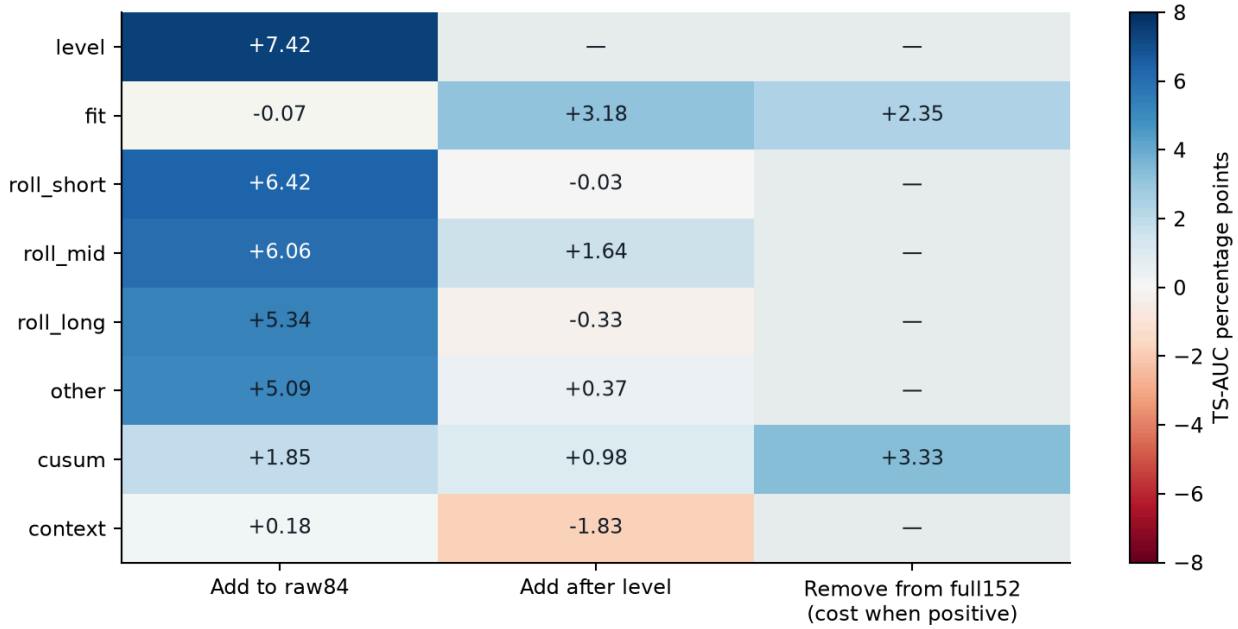


Figure 3. *Historical reduced-panel inclusion contrasts and available exclusion costs. Values are TS-AUC percentage points, positive when a family helps in the stated context. Gray cells are unmeasured. The exclusion baseline is the rounded full152 score 0.65281; the exact forward baseline is 0.55169576. No confidence interval or final-selection claim is attached to these reused-panel contrasts. The second column tests conditional additions after level features, demonstrating why forward ranking changes with the core.*

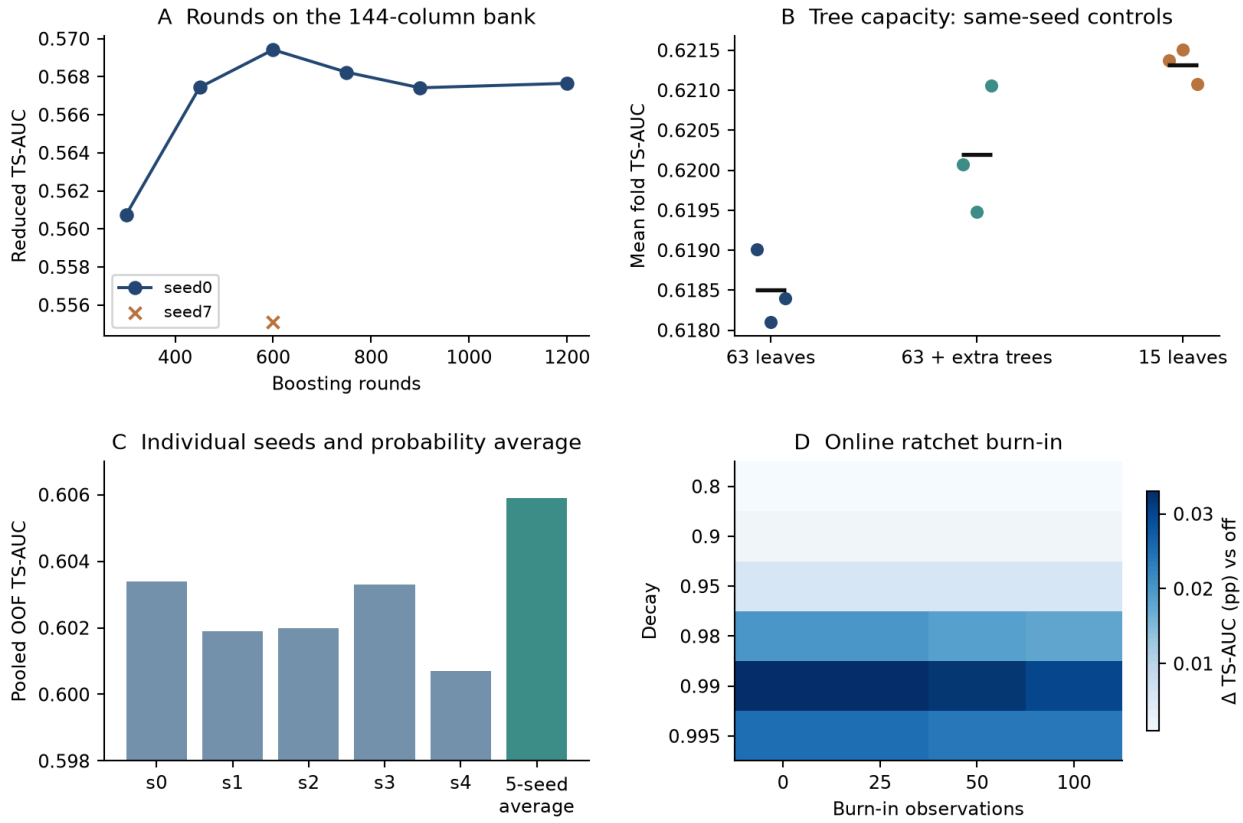


Figure 4. *Four historical diagnostics from saved experiments, each with its own control. A: reduced-panel rounds; seed7 at 600 illustrates variability, without an uncertainty estimate from just two points. B: seed0/1/2 means over the same five folds for the soft-student capacity screen; horizontal marks average these three seeds. This source saved predictions for all cells but its gate log ended after the first comparison, so it is not evidence that every gate completed. C: full-row BIC-AR-Rosenblatt individual early-stopped models versus their probability average; log values are rounded and the vertical axis is truncated. Averaging is distinct from choosing the highest-scoring seed. D: LightGBM/TCN blend online smoothing on 5,036,517 rows; the best observed ratchet gain is small and gates remain 0/5. Burn-in delays smoothing, not neural training. The panels cannot be compared as one matched model competition.*

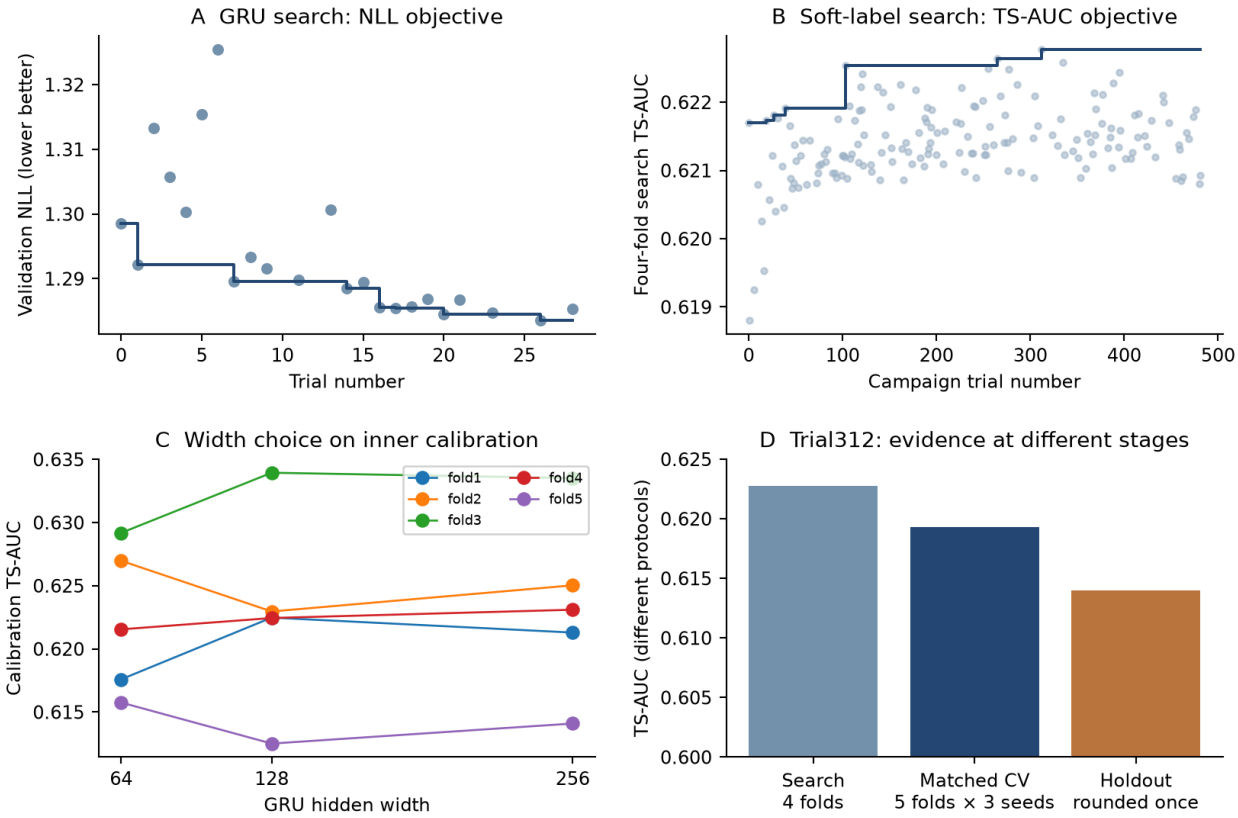


Figure 5. *A/B: completed Optuna trials and running best scores; pruned trials have no full-fold score in panel B. GRU NLL and soft-label TS-AUC are distinct objectives. C: inner-calibration width scores, frozen before each outer evaluation. D: trial312's selected search result, later matched validation and once-evaluated fixed holdout. These stages differ in folds, controls and score precision; their difference is not an unbiased numerical estimate of selection optimism. Search scores, reduced diagnostics and public submission scores are kept separate throughout the report.*

Appendix D GAS-t GPU global-search pilot

The eight search histories are 0, 1, 2, 3, 4, 5, 6, 7. The GPU smoke check matched the original CPU likelihood within 2.33×10^{-10}; all 56 saved solutions were checked again on their full observed histories, with maximum CPU/GPU difference 1.02×10^{-10}. Both methods use 128 candidates × 101 population evaluations (initial population plus 100 generations), with the same local anchor and parameter bounds. L-BFGS-B polish is additional, CPU-based work; its objective evaluations are recorded separately in the total count. A post-run diagnostic applies the same local polish directly to the saved anchor, separating improved local optimization from population-search contribution. A completed pilot is an optimizer diagnostic, not new detection validation.

Method	Mean NLL gain vs bounded local anchor	Mean NLL gain vs locally polished anchor	Additional seconds/history/seed
DE + local polish	+1.059531	+0.236212	1.81
PSO + local polish	+0.935852	+0.112532	0.94

Table D1. *CPU-verified full-history likelihood results, three search seeds on eight deliberately reused histories. The common four-start CPU anchor costs a further mean 17.33 seconds/history and is excluded from the additional run times. Timing includes transfers and local polishing, not compilation/loading. Lower NLL concerns history fit; this eight-history pilot performed no holdout, break-score, acceptance-gate or public-score evaluation. The later full-search detection test is reported in Section 5.5. The recorded holdout ledger remains unchanged.*

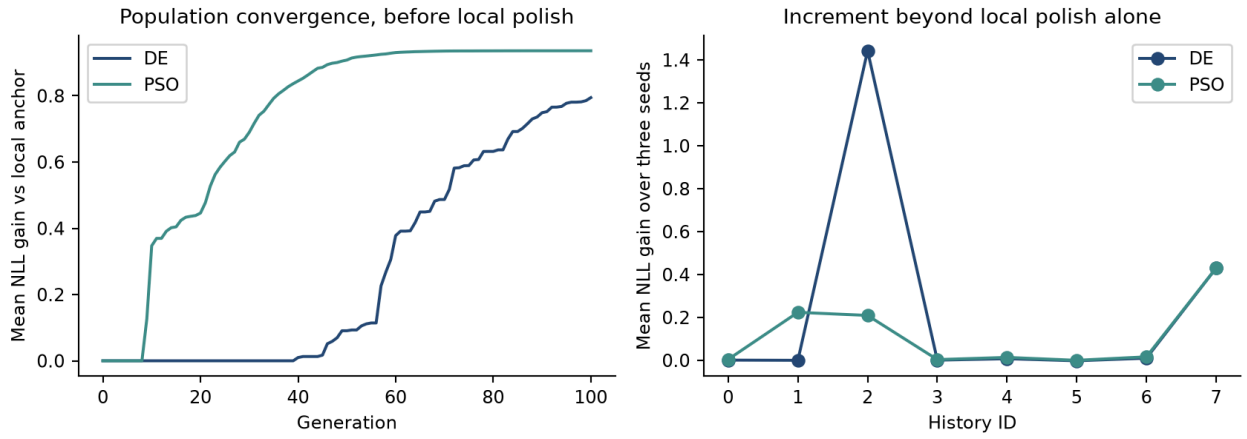


Figure 6. *Population best-NLL trajectories before local polishing, averaged across the same eight histories and three seeds; the right panel shows the final increment over applying local polish alone to the anchor. Every population includes the bounded local anchor, so retaining at least its unpolished fit is built into the protocol; this is not a fair speed race against an unseeded optimizer. Seed variation and finite-budget convergence do not certify global optimality. Improved history likelihood may overfit or produce less useful break residuals, with the corresponding full-search downstream test reported separately in Sections 5.5 and 6.1.*

Local polish alone improves the anchor by mean 0.823319 NLL. Consequently, most of the combined gain is local refinement. Population search adds a smaller mean increment: DE +0.236212, PSO +0.112532. DE does not beat the locally polished control in every individual seed/history cell. This post-run comparator is diagnostic, with no new data or acceptance claim.